

# Contents

|          |                                             |           |
|----------|---------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                         | <b>2</b>  |
| 1.1      | Arithmetic in $[0, \infty]$ . . . . .       | 2         |
| 1.2      | Metric Space . . . . .                      | 2         |
| 1.3      | Limits of Set Sequences . . . . .           | 3         |
| 1.4      | Exercises . . . . .                         | 6         |
| <b>2</b> | <b>Abstract Integration</b>                 | <b>9</b>  |
| 2.1      | The Concept of Measurability . . . . .      | 9         |
| 2.2      | Simple Functions . . . . .                  | 12        |
| 2.3      | Measure . . . . .                           | 13        |
| 2.4      | Integration of Positive Functions . . . . . | 15        |
|          | <b>Index</b>                                | <b>18</b> |

# Introduction

## 1.1 Arithmetic in $[0, \infty]$

## 1.2 Metric Space

### Lemma 1.2.1

$X$  is a metric space,  $x \in X$ ,  $B_1, B_2$  are two balls in  $X$ . If  $x \in B_1 \cap B_2$ , then  $x$  is the center of an open ball  $B \subseteq B_1 \cap B_2$ .

*Proof.* We may assume that  $B_1, B_2$  have different center. Let  $y_1 \neq y_2$ ,

$$B_1 = \{x : d(x, y_1) < r_1\}, \quad B_2 = \{x : d(x, y_2) < r_2\}$$

Take  $x \in B_1 \cap B_2$ , we claim that there exists  $r_0 > 0$  such that  $B := \{y : d(x, y) < r_0\} \subseteq B_1 \cap B_2$ . Otherwise, for all  $r > 0$ , there exists a point  $y_r$  such that the following two hold at the same time

1.  $d(x, y_r) < r$
2.  $d(y_r, y_1) \geq r_1$  or  $d(y_r, y_2) \geq r_2$ .

Thus

$$d(x, y_1) \geq d(y_r, y_1) - d(x, y_r) > r_1 - r, \quad \text{or} \quad d(x, y_2) > r_2 - r$$

The above shows that

$$r > \min\{r_1 - d(x, y_1), r_2 - d(x, y_2)\}, \quad \forall r > 0$$

which is a contradiction. □

## 1.3 Limits of Set Sequences

### Definition 1.3.1 ► Limit Inferior and Limit Superior

Let  $\{A_n\}_{n=1}^{\infty}$  be a sequence of sets.

1. By the *Limit Inferior of the sequence*, we mean

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \{x : \exists N_0 \in \mathbb{N}, \forall n > N_0, x \in A_n\}$$

2. By the *Limit Superior of the sequence*, we mean

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \{x : \forall N \in \mathbb{N}, \exists n \geq N, x \in A_n\}$$

*Note.*

1. Limit Inferior 包含那些“最终稳定下来”的元素，即从某个点之后就永远属于序列中所有后续集合的元素。
2. Limit Superior 包含那些“反复出现”的元素，即在无限多个集合中出现的元素。

### Proposition 1.3.2

For any sequence of sets  $\{A_n\}$ , it always holds that:

$$\liminf_{n \rightarrow \infty} A_n \subseteq \limsup_{n \rightarrow \infty} A_n$$

*Proof sketch.*

直觉上是显然的，因为“最终稳定下来”的元素一定会“反复出现”。

*Proof.* It is obvious by the  $\{x : x \in P\}$  form representation of the Limit Inferior and Superior. □

### Definition 1.3.3 ► Limit of a Set Sequence

If the limit inferior and limit superior of a set sequence  $\{A_n\}_{n=1}^{\infty}$  are equal, i.e.,  $\liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n$ , then we say the *limit* of the se-

quence exists, and it is defined as:

$$\lim_{n \rightarrow \infty} A_n = \liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n$$

### Proposition 1.3.4

Let  $\{A_n\}_{n=1}^{\infty}$  be a sequence of sets.

1. If  $\{A_n\}$  is an increasing sequence, then the limit of  $\{A_n\}$  exists and

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

2. If  $\{A_n\}$  is a decreasing sequence, then the limit of  $\{A_n\}$  exists and

$$\lim_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

*Proof.* 1. If  $\{A_n\}$  is increasing, then

$$\bigcap_{n=N}^{\infty} A_n = A_N$$

and

$$\bigcup_{n=N}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n, \forall N \in \mathbb{N}.$$

Thus

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} A_N$$

and

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

which are the same.

2. If  $\{A_n\}$  is decreasing, then

$$\bigcap_{n=N}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\bigcup_{n=N}^{\infty} A_n = A_N$$

Thus

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} A_N$$

which are the same.

□

### Proposition 1.3.5

For a sequence of sets  $\{A_n\}_{n=1}^{\infty}$  and their corresponding sequence of indicator functions  $\{\chi_{A_n}(x)\}_{n=1}^{\infty}$ :

- $\chi_{\liminf_{n \rightarrow \infty} A_n}(x) = \liminf_{n \rightarrow \infty} \chi_{A_n}(x)$
- $\chi_{\limsup_{n \rightarrow \infty} A_n}(x) = \limsup_{n \rightarrow \infty} \chi_{A_n}(x)$
- $\lim_{n \rightarrow \infty} A_n$  exists, then  $\chi_{\lim_{n \rightarrow \infty} A_n}(x) = \lim_{n \rightarrow \infty} \chi_{A_n}(x)$

*Proof sketch.*

- 若  $x$  在 limit inferior 里面, 则  $x$  是“最终稳定”的,  $\chi_{A_n}(x)$  是关于  $n$  “最终”恒为 1 的.
- 若  $x$  在 limit superior 里面, 则  $x$  是“反复出现”的, 即相当于  $N$  多大, 总会出现之后的某个  $n$  使得  $\chi_{A_n}(x) = 1$ .
- 当极限存在时, 函数列  $\{\chi_{A_n}(x)\}_{n=1}^{\infty}$  的 limit inferior 和 supperior 根据上两条相等, 等于极限集合的  $\chi$ .

## 1.4 Exercises

### Problem 1.4.1 ► Convex Bodies

A convex body is a compact, convex subset  $K \subseteq \mathbb{R}^n$  which contains 0 as an interior point and which is symmetric around the origin .

1. Prove that the following formula defines a norm on  $\mathbb{R}^n$ :

$$N(x) := \inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\}$$

2. Prove that  $K$  is homeomorphic to  $\mathbb{D}^n$  and that  $\partial K$  is homeomorphic to  $\mathbb{S}^{n-1}$  .

*Note.*

1. 若  $K$  是一个单位球. 则  $N(x) = \|x\|$ .
2. 证明  $N(x)$  是一个 norm, 需要证明: 1. 非负正定 2. 绝对一次齐次 3. 三角不等式.
3. 由于  $\varphi$  的形式足够简单, 可以通过直接构造逆映射证明双射.

*Proof.* 1. It is obvious that  $N(x) \geq 0$ . If  $N(x) = 0$  , then

$$\inf \left\{ \lambda > 0 : \frac{x}{\lambda} \in K \right\} = 0$$

which means that for all  $\varepsilon > 0$ ,  $\frac{x}{\varepsilon} \in K$ . Since  $K$  is compact,  $\text{dist}(K, 0) < \infty$ . If  $x \neq 0$ , then  $\|x\| > 0$ . We take  $\varepsilon_0 < \frac{1}{\text{dist}(K, 0) \cdot \|x\|}$ . Then  $\left\| \frac{x}{\varepsilon_0} \right\| > \text{dist}(K, 0)$ . But  $\frac{x}{\varepsilon_0} \in K$ , which is a contradiction. For the second part, let  $x \in \mathbb{R}^n, k \in \mathbb{R} \setminus \{0\}$ . For all  $\lambda > N(x)$ ,

$$\frac{x}{\lambda} \in K \implies \frac{kx}{k\lambda} \in K$$

, which shows that  $N(kx) \leq kN(x)$ . Replace  $k$  by  $\frac{1}{k}$  and replace  $x$  by  $kx$ , we have  $N(kx) \geq kN(x)$ . Thus  $N(kx) = kN(x)$ . For the triangle inequality, we need to show that

$$\inf \left\{ \lambda > 0 : \frac{x_1 + x_2}{\lambda} \in K \right\} \leq \inf \left\{ \lambda > 0 : \frac{x_1}{\lambda} \in K \right\} + \inf \left\{ \lambda > 0 : \frac{x_2}{\lambda} \in K \right\}$$

Let  $\lambda_0 = N(x_1 + x_2)$ . Only need to show that

$$\frac{x_1 + x_2}{N(x_1) + N(x_2)} \in K$$

For all  $\delta > 0$ , we have

$$\frac{x_1}{N(x_1) + \delta} \in K, \quad \frac{x_2}{N(x_2) + \delta} \in K$$

. Since  $K$  is convex, we have the convex combination

$$\frac{1}{N(x_1) + N(x_2) + 2\delta} \left( (N(x_2) + \delta) \frac{x_2}{N(x_2) + \delta} + (N(x_1) + \delta) \frac{x_1}{N(x_1) + \delta} \right) \in K$$

That is

$$\frac{x_1 + x_2}{N(x_1) + N(x_2) + 2\delta} \in K$$

for all  $\delta > 0$ . Then

$$N(x_1 + x_2) \leq N(x_1) + N(x_2) + 2\delta, \quad \forall \delta > 0 \implies N(x_1 + x_2) \leq N(x_1) + N(x_2)$$

2.  $x \in K^\circ \iff N(x) < 1$ ,  $x \in \text{Ext}(K) \iff N(x) > 1$ ,  $x \in \partial K \iff N(x) = 1$ . We define a map  $\varphi : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^n$ .

$$\varphi(x) := \begin{cases} N(x) \frac{x}{\|x\|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

We have

$$\|\varphi(x)\| = N(x)$$

Then  $\varphi$  maps  $K^\circ$  on to  $(\mathbb{D}^n)^\circ$ , and maps  $\partial K$  on to  $\mathbb{S}^{n-1}$ . To show  $\varphi$  is continuous, only need to show it is continuous at 0. In fact,  $\|\lim_{x \rightarrow 0} \varphi(x)\| = \lim_{x \rightarrow 0} \|\varphi(x)\| = \lim_{x \rightarrow 0} N(x) = 0$  since  $N$  is a norm. We define  $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ :

$$\psi(x) := \begin{cases} \frac{\|x\|}{N(x)} x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then  $\varphi \circ \psi = \psi \circ \varphi = \text{Id}$ . The same skills can be used to show  $\psi$  is continuous at 0 as well. Finally, since  $\mathbb{S}^{n-1}$  is a subspace of  $\mathbb{D}^n$ ,  $\psi|_{\mathbb{S}^{n-1}} : \mathbb{S}^{n-1} \rightarrow \psi(\mathbb{S}^{n-1}) = \partial K$  is also a homeomorphism.

□

**Problem 1.4.2**

Let  $X, X'$  be topological spaces and  $x \in X, x' \in X'$  be base points. Prove that the wedge sum  $X \vee X'$  is homeomorphic to the subspace:

$$(X \times \{x'\}) \cup (\{x\} \times X') \subseteq X \times X'$$



# Abstract Integration

## 2.1 The Concept of Measurability

### Definition 2.1.1

1. A collection  $\mathfrak{M}$  of subsets of a set  $X$  is said to be a  $\sigma$ -**algebra** in  $X$  if  $\mathfrak{M}$  has the following properties:
  - (a)  $X \in \mathfrak{M}$ .
  - (b) If  $A \in \mathfrak{M}$ , then  $A^c \in \mathfrak{M}$ .
  - (c) If  $A = \bigcup_{n=1}^{\infty} A_n$  and  $A_n \in \mathfrak{M}$  for  $n = 1, 2, 3, \dots$ , then  $A \in \mathfrak{M}$ .
2. If  $\mathfrak{M}$  is a  $\sigma$ -algebra in  $X$ , then  $X$  is called a **measurable** provided that  $f^{-1}(V)$  is measurable set in  $X$  for every open set  $V$  in  $Y$ .
3. If  $X$  is a measurable space,  $Y$  is a topological space, and  $f$  is a mapping of  $X$  into  $Y$ , then  $f$  is said to be **measurable** provided that  $f^{-1}(V)$  is a measurable set in  $X$  for every open set  $V$  in  $Y$ .

*Note.*

对比 topological space 的定义,  $\sigma$ -algebra 的定义是对称的, 即可测集的补集仍是可测的, 而对于拓扑空间, 开集的补集不是开集, 而是被定义为了闭集这样的对象.

### Lemma 2.1.2

1. Let  $X$  be a measurable space,  $Y, Z$  be topological spaces. If  $f : X \rightarrow Y$  is measurable, and if  $g : Y \rightarrow Z$  is continuous, then  $g \circ f : X \rightarrow Z$  is measurable.
2. Let  $u, v$  be real measurable functions on a measurable space  $X$ , let  $\Phi$  be a continuous mapping of the plane into a topological space  $Y$ , and define

$$h(x) = \Phi(u(x), v(x))$$

for  $x \in X$ . Then  $h : X \rightarrow Y$  is measurable.

**Corollary 2.1.3**

Let  $X$  be a measurable space. The following propositions hold

1. If  $f = u + iv$ , where  $u$  and  $v$  are real measurable functions on  $X$ , then  $f$  is a complex measurable function on  $X$ .
2. If  $f = u + iv$  is a complex measurable function on  $X$ , then  $u$ ,  $v$ , and  $|f|$  are real measurable functions on  $X$ .
3. If  $f$  and  $g$  are complex measurable functions on  $X$ , then so are  $f + g$  and  $fg$ .
4. If  $E$  is a measurable set in  $X$  and if

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

then  $\chi_E$  is a measurable function.

5. If  $f$  is a complex measurable function on  $X$ , there is a complex measurable function  $\alpha$  on  $X$  such that  $|\alpha| = 1$  and  $f = \alpha |f|$ .

**Theorem 2.1.4**

If  $\mathcal{F}$  is any collection of subsets of  $X$ , there exists a smallest  $\sigma$ -algebra  $\mathfrak{M}^*$  such that  $\mathcal{F} \subseteq \mathfrak{M}^*$ .

**Definition 2.1.5 ▶ Borel**

Let  $X$  be a topological space.

1. By **Borel  $\sigma$ -algebra**, we mean the smallest  $\sigma$ -algebra  $\mathcal{B}$  in  $X$  such that every open set in  $X$  belongs to  $\mathcal{B}$ . The members of  $\mathcal{B}$  are called the **Borel sets** of  $X$ .
2. All countable unions of closed sets and all countable intersections of open sets are Borel sets, which we called  $F_\sigma$ 's **and**  $G_\delta$ 's, respectively.
3. By a **Borel function**, we mean a measurable function on the measurable space  $(X, \mathcal{B})$ .

*Remark.*

1. The letters  $F$  and  $G$  were used for closed and open sets, respectively, and  $\sigma$  refers to union,  $\delta$  to intersection.
2. A continuous function is Borel-measurable, since the preimage of any open set is open and therefore a Borel set.

**Theorem 2.1.6**

Suppose  $\mathfrak{M}$  is a  $\sigma$ -algebra in  $X$ , and  $Y$  is a topological space. Let  $f$  map  $X$  into  $Y$ .

1. If  $\Omega$  is the collection of all sets  $E \subseteq Y$  such that  $f^{-1}(E) \in \mathfrak{M}$ , then  $\Omega$  is a  $\sigma$ -algebra in  $Y$ .
2. If  $f$  is measurable and  $E$  is a Borel set in  $Y$ , then  $f^{-1}(E) \in \mathfrak{M}$ .
3. If  $Y = [-\infty, \infty]$  and  $f^{-1}((\alpha, \infty]) \in \mathfrak{M}$  for every real  $\alpha$ , then  $f$  is measurable.
4. If  $f$  is measurable, if  $Z$  is topological space, if  $g : Y \rightarrow Z$  is a Borel mapping, and if  $h = g \circ f$ , then  $h : X \rightarrow Z$  is measurable.

**Theorem 2.1.7**

If  $f_n : X \rightarrow [-\infty, \infty]$  is measurable, for  $n = 1, 2, 3, \dots$ , and

$$g = \sup_{n \geq 1} f_n, \quad h = \lim_{n \rightarrow \infty} \sup f_n,$$

then  $g$  and  $h$  are measurable.

**Corollary 2.1.8**

1. The limit of every pointwise convergent sequence of complex measurable functions is measurable.
2. If  $f$  and  $g$  are measurable (with range in  $[-\infty, \infty]$ ), then so are  $\max\{f, g\}$  and  $\min\{f, g\}$ . In particular, this is true of the functions

$$f^+ = \max\{f, 0\}, \quad f^- = -\min\{f, 0\}.$$

which are called the **positive part** and **negative part** of  $f$ , respectively.

*Remark.*

1. There are the standard representation

$$|f| = f^+ + f^-, \quad f = f^+ - f^-$$

2. And easy but (may) useful observation is: If  $f = g - h$ ,  $g \geq 0$ ,  $h \geq 0$ , then  $f^+ \leq g$  and  $f^- \leq h$ .

## 2.2 Simple Functions

### Definition 2.2.1

A complex function  $s$  on a measurable space  $X$  whose range consists of only finitely many points will be called a **simple function**. Among these are the nonnegative simple functions, whose range is a finite subset of  $[0, \infty)$ .

Specifically, if  $\alpha_1, \dots, \alpha_n$  are distinct values of a simple function  $s$ , and if we set  $A_i = \{x : s(x) = \alpha_i\}$ , then clearly

$$s = \sum_{i=1}^n \alpha_i \chi_{A_i}.$$

Where  $\chi_{A_i}$  is the characteristic function of  $A_i$ .

*Remark.*

- Here, we explicitly exclude  $\infty$  from the values of a simple function.
- It is clear that  $s$  is measurable if and only if each of the sets  $A_i$  is measurable.

### Theorem 2.2.2

Let  $f : X \rightarrow [0, \infty]$  be measurable space. There exists simple measurable functions  $s_n$  on  $X$  such that

1.  $0 \leq s_1 \leq s_2 \leq \dots \leq f$ .
2.  $s_n(x) \rightarrow f(x)$  as  $n \rightarrow \infty$ , for every  $x \in X$ .

*Proof sketch.*

We construct a sequence of Borel simple functions  $\{\varphi_n(x)\}$  to act as an **identity**. 当  $n$  增大的同时, 我们同时让  $\varphi_n(x)$  的单位逼近精度和单位逼近范围随着  $n$  提升. 并且在舍弃误差时, 总是向下取整, 使得该单位逼近是自下而上的.

*Proof.* For every  $x \in [0, \infty]$ , and for every  $n \in \mathbb{N}$ , there exists a unique integer  $k_n(x)$ , such that

$$k_n(x) 2^{-n} \leq x < (k_n(x) + 1) 2^{-n}$$

For every  $n \in \mathbb{N}$ , we define

$$\varphi_n(x) = \begin{cases} k_n(x) 2^{-n}, & 0 \leq x \leq n \\ n, & x \geq n \end{cases}$$

Each  $\varphi_n(x)$  is then a Borel simple function. It is not hard to show that  $0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq \text{Id}$ . For each  $n$ , we define

$$s_n(x) := (\varphi_n \circ f)(x)$$

Then  $\{s_n\}$  is a suquence of simple measurable functions such that  $0 \leq s_1 \leq s_2 \leq \dots \leq f$ . Since  $\lim_{n \rightarrow \infty} \varphi_n(x) = x$ , then  $\lim_{n \rightarrow \infty} s_n(x) = f(x)$ .  $\square$

## 2.3 Measure

### Definition 2.3.1 ► Measure and Measure Space

- (a) A **positive measure** is a function  $\mu$ , defined on a  $\sigma$ -algebra  $\mathfrak{M}$ , whose range is in  $[0, \infty]$  and which is **countably additive**. This means that if  $\{A_i\}$  is a disjoint countable collection of members of  $\mathfrak{M}$ , then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

To avoid trivialities, we shall also assume that  $\mu(A) < \infty$  for at least one  $A \in \mathfrak{M}$ .

- (b) A **measure space** is a measurable space which has a positive measure defined on the  $\sigma$ -algebra of its measurable sets.
- (c) A **complex measure** is a complex-valued countably additive function defined on a  $\sigma$ -algebra.

### Theorem 2.3.2

Let  $\mu$  be a positive measure on a  $\sigma$ -algebra  $\mathfrak{M}$ . Then

1.  $\mu(\emptyset) = 0$ .
2.  $\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n)$  if  $A_1, \dots, A_n$  are pairwise disjoint members of  $\mathfrak{M}$ .
3.  $A \subset B$  implies  $\mu(A) \leq \mu(B)$  if  $A \in \mathfrak{M}, B \in \mathfrak{M}$ .

4.  $\mu(A_n) \rightarrow \mu(A)$  as  $n \rightarrow \infty$  if  $A = \bigcup_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$ , and

$$A_1 \subset A_2 \subset A_3 \subset \dots$$

5.  $\mu(A_n) \rightarrow \mu(A)$  as  $n \rightarrow \infty$  if  $A = \bigcap_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$ ,

$$A_1 \supset A_2 \supset A_3 \supset \dots,$$

and  $\mu(A_1)$  is finite.

*Proof.* 1. Take  $A \in \mathfrak{M}$  such that  $\mu(A) < \infty$ .<sup>1</sup> And let  $A_2 = A_3 = \dots = \emptyset$ , then  $\mu(\emptyset) > 0$  leads to a contradiction to the countably additive.

2. Take  $A_{n+1} = A_{n+2} = \dots = \emptyset$ .

3. Note that  $B = (B \setminus A) \cup A$ , then by additivity

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

4. Let  $A_0 = \emptyset$ , and let  $B_n = A_n \setminus A_{n-1}$  for all  $n \in \mathbb{N}$ . Then  $B_1, \dots, B_n$  are pairwise disjoint members of  $\mathfrak{M}$  such that  $A = \bigcup_{n=1}^{\infty} B_n$ . We have

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1})$$

If one of the  $\mu(A_n)$  is  $\infty$ , then  $\lim_{n \rightarrow \infty} \mu(A_n)$  and  $\mu(A)$  both are  $\infty$ . Otherwise, we have

$$\mu(A_n \setminus A_{n-1}) = \mu(A_n) - \mu(A_{n-1}), \quad \forall n \in \mathbb{N}$$

Thus

$$\mu(A) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1}) = \lim_{n \rightarrow \infty} \mu(A_n)$$

5. Let  $B_n = A_n \setminus A_{n+1}$  for all  $n \in \mathbb{N}$ .  $B_1, \dots, B_n$  are pairwise disjoint members of  $\mathfrak{M}$  with finite measure, such that

$$A_1 \setminus A = A_1 \setminus \left( \bigcap_{n=1}^{\infty} A_n \right) = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = \bigcup_{n=1}^{\infty} \left( \bigcup_{k=1}^n B_k \right) = \bigcup_{n=1}^{\infty} B_n$$

<sup>1</sup>That is what we supposed at the definition of measure.

. We have

$$\mu(A_1 \setminus A) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right),$$

where the RHS is  $\mu(A_1) - \mu(A)$ , and the LHS is

$$\sum_{n=1}^{\infty} (\mu(A_n) - \mu(A_{n+1})) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_{n+1})$$

Since  $\mu(A_1) < \infty$ , we have

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n)$$

□

### Example 2.3.3

1. For any  $E \subset X$ , where  $X$  is any set, define  $\mu(E) = \infty$  if  $E$  is an infinite set, and let  $\mu(E)$  be the number of points in  $E$  if  $E$  is finite. This  $\mu$  is called the **counting measure** on  $X$ .
2. Fix  $x_0 \in X$ , define  $\mu(E) = 1$  if  $x_0 \in E$  and  $\mu(E) = 0$  if  $x_0 \notin E$ , for any  $E \subset X$ . This  $\mu$  may be called the **unit mass concentrated at  $x_0$** .
3. Let  $\mu$  be the counting measure on the set  $\{1, 2, 3, \dots\}$ , let  $A_n = \{n, n+1, n+2, \dots\}$ . Then  $\bigcap A_n = \emptyset$  but  $\mu(A_n) = \infty$  for  $n = 1, 2, 3, \dots$ . This shows that the hypothesis

$$\mu(A_1) < \infty$$

is not superfluous in Theorem 2.3.2(5).

## 2.4 Integration of Positive Functions

### Definition 2.4.1 ► Integration of Positive Functions

1. If  $s : X \rightarrow [0, \infty)$  is a measurable simple function, of the form

$$s = \sum_{i=1}^n \alpha_i \chi_{A_i},$$

where  $\alpha_1, \dots, \alpha_n$  are the distinct values of  $s$ , and if  $E \in \mathfrak{M}$ , we define

$$\int_E s \, d\mu = \sum_{i=1}^n \alpha_i \mu(A_i \cap E).$$

The convention  $0 \cdot \infty = 0$  is used here; it may happen that  $\alpha_i = 0$  for some  $i$  and that  $\mu(A_i \cap E) = \infty$ .

2. If  $f : X \rightarrow [0, \infty]$  is measurable, and  $E \in \mathfrak{M}$ , we define

$$\int_E f \, d\mu = \sup \int_E s \, d\mu,$$

the supremum being taken over all simple measurable functions  $s$  such that  $0 \leq s \leq f$ . The left member above is called the **Lebesgue integral** of  $f$  over  $E$ , with respect to the measure  $\mu$ . It is a number in  $[0, \infty]$ .

*Remark.*

We apparently have two definitions for  $\int_E f \, d\mu$  if  $f$  is simple, they are the same.

### Corollary 2.4.2

- (a) If  $0 \leq f \leq g$ , then  $\int_E f \, d\mu \leq \int_E g \, d\mu$ .
- (b) If  $A \subset B$  and  $f \geq 0$ , then  $\int_A f \, d\mu \leq \int_B f \, d\mu$ .
- (c) If  $f \geq 0$  and  $c$  is a constant,  $0 \leq c < \infty$ , then

$$\int_E cf \, d\mu = c \int_E f \, d\mu.$$

- (d) If  $f(x) = 0$  for all  $x \in E$ , then  $\int_E f \, d\mu = 0$ , even if  $\mu(E) = \infty$ .
- (e) If  $\mu(E) = 0$ , then  $\int_E f \, d\mu = 0$ , even if  $f(x) = \infty$  for every  $x \in E$ .
- (f) If  $f \geq 0$ , then  $\int_E f \, d\mu = \int_X \chi_E f \, d\mu$ .

*Remark.*

We can also regard the  $\int_E f \, d\mu$  as the restricted function  $f|_E$  integrates on the induced measure space  $E$ .



**Proposition 2.4.3**

Let  $s$  and  $t$  be nonnegative measurable simple functions on  $X$ . For  $E \in \mathfrak{M}$ , define

$$\varphi(E) = \int_E s \, d\mu.$$

Then  $\varphi$  is a measure on  $\mathfrak{M}$ . Also

$$\int_X (s + t) \, d\mu = \int_X s \, d\mu + \int_X t \, d\mu.$$

*Proof.* If  $E_1, E_2, \dots$  are disjoint members of  $\mathfrak{M}$  whose union is  $E$ , the countable additivity of  $\mu$  shows that

$$\begin{aligned} \varphi(E) &= \sum_{i=1}^n \alpha_i \mu(A_i \cap E) \\ &= \sum_{i=1}^n \sum_{j=0}^{\infty} \alpha_i \mu(A_i \cap E_j) \\ &= \sum_{j=0}^{\infty} \sum_{i=1}^n \alpha_i \mu(A_i \cap E_j) \\ &= \sum_{j=0}^{\infty} \varphi(E_j) \end{aligned}$$

Also,  $\varphi(\emptyset) \neq \infty$ ,  $\mu$  is not identically  $\infty$ .

Next, let  $\beta_1, \dots, \beta_m$  be distinct values of  $t$ . And let  $B_j := \{x : t(x) = \beta_j\}$ . Then

$$\begin{aligned} \int_X (s + t) \, d\mu &= \sum_{i,j} \int_{A_i \cap B_j} (s + t) \, d\mu \\ &= \sum_{i,j} \alpha_i \mu(A_i \cap B_j) + \sum_{i,j} \beta_j \mu(A_i \cap B_j) \\ &= \sum_i \alpha_i \mu(A_i) + \sum_j \beta_j \mu(B_j) \\ &= \int_X s \, d\mu + \int_X t \, d\mu \end{aligned}$$

□

**Theorem 2.4.4 ▶ Lebesgue's Monotone Convergence Theorem**

Let  $\{f_n\}$  be a sequence of measurable functions on  $X$ , and suppose that

(a)  $0 \leq f_1(x) \leq f_2(x) \leq \dots \leq \infty$  for every  $x \in X$ ,

(b)  $f_n(x) \rightarrow f(x)$  as  $n \rightarrow \infty$ , for every  $x \in X$ .

Then  $f$  is measurable, and

$$\int_X f_n d\mu \rightarrow \int_X f d\mu \quad \text{as } n \rightarrow \infty.$$

*Proof.* Since  $\int_X f_n d\mu \leq \int_X f_{n+1} d\mu$  for all  $n$ , there exists an  $\alpha > 0$  such that

$$\int_X f_n d\mu \rightarrow \alpha, \quad \text{as } n \rightarrow \infty$$

By Theorem 2.1.7,  $f$  is measurable, then by  $\int_X f_n d\mu \leq \int_X f d\mu \forall n$ ,

$$\alpha \leq \int_X f d\mu$$

□

# Index

## Definitions

|       |                                                 |    |
|-------|-------------------------------------------------|----|
| 1.3.1 | Limit Inferior and<br>Limit Superior . . . .    | 3  |
| 1.3.3 | Limit of a Set Sequence                         | 3  |
| 2.1.1 | . . . . .                                       | 9  |
| 2.1.5 | Borel . . . . .                                 | 10 |
| 2.2.1 | . . . . .                                       | 12 |
| 2.3.1 | Measure and Mea-<br>sure Space . . . . .        | 13 |
| 2.4.1 | Integration of Posi-<br>tive Funtions . . . . . | 15 |

## Examples

|       |           |    |
|-------|-----------|----|
| 2.3.3 | . . . . . | 15 |
|-------|-----------|----|

## Problems

|       |                       |   |
|-------|-----------------------|---|
| 1.4.1 | Convex Bodies . . . . | 6 |
| 1.4.2 | . . . . .             | 8 |

## Propositions

|       |           |   |
|-------|-----------|---|
| 1.3.2 | . . . . . | 3 |
|-------|-----------|---|

|       |           |    |
|-------|-----------|----|
| 1.3.4 | . . . . . | 4  |
| 1.3.5 | . . . . . | 5  |
| 2.4.3 | . . . . . | 17 |

## Corollarys

|       |           |    |
|-------|-----------|----|
| 2.1.3 | . . . . . | 10 |
| 2.1.8 | . . . . . | 11 |
| 2.4.2 | . . . . . | 16 |

## Lemmas

|       |           |   |
|-------|-----------|---|
| 1.2.1 | . . . . . | 2 |
| 2.1.2 | . . . . . | 9 |

## Theorems

|       |                                                           |    |
|-------|-----------------------------------------------------------|----|
| 2.1.4 | . . . . .                                                 | 10 |
| 2.1.6 | . . . . .                                                 | 11 |
| 2.1.7 | . . . . .                                                 | 11 |
| 2.2.2 | . . . . .                                                 | 12 |
| 2.3.2 | . . . . .                                                 | 13 |
| 2.4.4 | Lebesgue's Mono-<br>tone Convergence<br>Theorem . . . . . | 18 |